Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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***Abstract*—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.**

I. Introduction

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. Related Work

Workflow-centred mitigations and verification tools. Surveys and recent system reports document architectures that pair LLM generation with validators, verifiers, and emulators to improve correctness of generated NetOps artifacts and adjacent program-generation tasks [1], [2]. Tool-focused work demonstrates generate→validate→repair patterns (e.g., Batfish/NetKAT style verification, emulator-assisted validation) but reports heterogeneous gains depending on task, validator rule coverage, and emulator fidelity [2], [3].

Empirical findings motivating a paired design. Empirical studies in program/specification generation show external verifiers often catch model mistakes not found by the model alone, and some NetOps prototypes use emulation to expose runtime issues [3]. These prior findings motivated our preregistered paired design that isolates guarded-stage contributions: because we enforce shared_initial_candidate semantics, the only permissible sources of between-condition discordance are downstream guarded-stage emulation observations and permitted repair-stage invocations recorded in run accounting.

III. Methodology

Preregistration and experimental design. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). Primary estimand: paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples. Planned confirmatory sample: Npairs = 720 (planned episodes = 1440). Minimum effect-of-interest for power planning = 0.20 (20 percentage points). The preregistration specified inclusion/exclusion rules, missingness handling, multiplicity handling for exploratory secondary tests, and deterministic contamination controls.

Task bank, vendors, and inclusion. Confirmatory items were synthetic intent→configuration tasks from intent_configuration_repair_v1 covering three abstracted vendor-CLI families and three topology templates (leaf-spine, 3-node service chain, triangle). Inclusion required vendor-specific parser acceptance; the preregistered missingness policy permitted up to two deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically excluded and replaced. In execution there were zero irrecoverable parser exclusions: complete_pair_count = 720 (see analysis/missingness_summary.csv).

Model configuration and sampling semantics. Declared/requested model: "gpt-5-mini" (provider field: openai_responses). The archived model configuration records max_output_tokens = 2000 and temperature = null (execution/model_configuration.json, sha256 a40e96e212ab87da...). Important clarification (preregistered and enforced): temperature = null indicates the client did not set an explicit temperature parameter; null does not imply temperature=0 or provider-side determinism. We enforced shared_initial_candidate semantics: for each pair we obtained exactly one sampled initial candidate (shared) which both baseline and guarded pipelines analyzed. Therefore any between-condition difference can only arise from (i) guarded-stage emulation observations not encoded as a validator rule, or (ii) the allowed downstream repair-stage invocation(s) recorded in run accounting.

Validator and guarded pipeline implementation. Baseline: deterministic_netops_validator_v5 performing full intent-constraint validation (syntax, command ordering/workflow, required-field presence, and policy-level constraints) and returning binary detected/not-detected per the preregistered scoring rule full_intent_constraint_validation. Guarded: baseline validator followed by an automated sandboxed-emulation harness executing the candidate against an emulated instance of the declared topology and producing runtime observations. The repair policy permitted at most one repair-stage model call per task; run accounting records exactly one repair-stage call in the confirmatory execution.

Execution controls, provenance, and deterministic reconciliation. The adapter family used: hosted_netops_gvr_v1. Execution manifest and deterministic reconciliation artifacts record completed_episode_count = 1440, complete_pair_count = 720, failed_episode_count = 0, and provider-call audit information including model_calls_used = 721 and stage_counts: shared_initial_generation = 720; repair = 1. Provenance and contamination controls (deterministic hashing, artifact-version checks) were applied per the preregistration; no contamination flags occurred for the confirmatory set.

IV. Results

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts (analysis/paired_contingency_table.csv and analysis/condition_summary.csv) report the contingency counts: n11 = 719 (detected by both), n10 = 1 (guarded-only), n01 = 0 (baseline-only), n00 = 0 (neither). Marginal detection rates: baseline = 719/720 = 0.9986111111111111; guarded = 720/720 = 1.0. Paired estimate (guarded - baseline) = 0.001388888888888884. Exact McNemar two-sided test p = 1.0. Paired-bootstrap percentile 95% CI (10000 resamples, bootstrap_seed = 1729) = [0.0, 0.004166666666666667]. All numeric values below are reproduced verbatim from analysis/results.json and analysis artifacts.

Table 1 — Primary confirmatory summary (artifact-grounded). - Npairs = 720 (complete paired episodes) - Baseline successes = 719; Baseline success rate = 0.9986111111111111 - Guarded successes = 720; Guarded success rate = 1.0 - Paired difference (guarded - baseline) = 0.001388888888888884 - Exact McNemar two-sided p = 1.0 - 95% bootstrap percentile CI = [0.0, 0.004166666666666667] (10000 resamples, seed = 1729)

Discordant-pair attribution and repair accounting. Provider-call accounting and deterministic reconciliation identify a single repair-stage invocation (stage_counts: repair = 1) and a unique n10 discordant count. Because shared_initial_candidate semantics were enforced, the permissible causes for that single guarded-only detection are: (i) a guarded-stage emulation observation not encoded as a validator violation, or (ii) the single recorded repair-stage invocation. The confirmatory execution bundle contains the raw per-episode records; the unique discordant record can be located by searching execution/raw_results.jsonl (input-results SHA-256 = df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a 602e0d2513aef0e8) for the single entry where baseline_score = 0 and guarded_score = 1. Deterministic reconciliation is archived as analysis/deterministic_reconciliation.json (sha256 49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a 42bcb096d0dec3aa) and its provider_call_audit confirms model_calls_used = 721 and repair = 1. The raw_results record for the discordant entry contains the pair_id and the validator_trace and repair_provider_trace fields that link the guarded detection to the documented repair-stage call; these artifacts are part of the canonical execution bundle referenced by the execution manifest.

Representative per-episode diagnostics. Representative scoring traces archived under execution/scoring/task-000001-*.json ... task-000006-*.json include validator_trace objects with normalized_configuration, validation_before/after, parsed_command_count, required_command_count, violation_count, and validator_version = 5.0. Those representative artifacts illustrate that the vast majority of shared_initial candidates complied with validator rules (violation_count = 0) and that sandboxed emulation largely corroborated validator findings in this curated task bank.

V. Discussion

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors and suggests practical boundary conditions where guarded stages are—and are not—likely to matter.

1) Validator ceiling effect and task-rule alignment. Deterministic_netops_validator_v5 explicitly encodes the syntactic and semantic checks exercised by the curated tasks (examples of patterns in task payloads include shutdown_configure_restore, make_before_break_uplink_switchover, and

paired_link_atomic_mtu_change). Representative validator_trace fields (parsed_command_count, required_command_count, violation_count = 0) in execution/scoring/task-000001-*.json show that most shared_initial candidates already satisfied the validator's rule set. The baseline marginal success rate of 719/720 (0.9986111111111111) leaves essentially no headroom for the guarded emulation stage to flag additional failures in this constrained regime.

2) Curated inclusion and reduced adversarial exposure. The confirmatory sample was assembled from a deterministic generator (intent_configuration_repair_v1) and required vendor-parser acceptance before inclusion. This inclusion rule intentionally restricted confirmatory items to parser-validated, policy-constrained scenarios; as a result, many runtime-only faults that emulation detects in more diverse or adversarial settings are underrepresented. The missingness summary and deterministic_reconciliation artifacts confirm zero irrecoverable parser exclusions, indicating the set was tightly constrained to inputs the validator could reasonably analyze.

3) Shared_initial_candidate semantics and limited repair activity. By design we reused the exact sampled initial candidate for both baseline and guarded conditions; this isolates downstream guarded-stage signals as the only allowable source of between-condition discordance. Provider_call_audit and deterministic_reconciliation report model_calls_used = 721 and stage_counts: shared_initial_generation = 720, repair = 1. The single n10 discordant pair therefore corresponds to the single recorded repair-stage invocation or to a guarded-emulation observation not codified as a validator rule. The raw_results and deterministic_reconciliation artifacts together allow an auditor to locate that unique pair_id and inspect validator_trace and repair_provider_trace to see whether the repair altered the candidate or whether the emulation exposed a runtime-only violation.

Interpretation and operational implications. For environments that maintain a high-coverage static validator tailored to expected intent patterns, and where inputs are parser-validated and curated, adding an automated sandboxed-emulation stage may have negligible incremental detection benefit at scale. This does not imply sandboxing is unnecessary; rather, it implies a conditional efficiency boundary: when validator coverage closely matches expected task patterns and the input pipeline enforces strong prevalidation, guarded runtime checks will detect relatively few additional faults. Conversely, in operational regimes characterized by (a) noisy natural-language intents, (b) adversarially crafted misconfigurations, (c) broader uncurated vendor-CLI diversity, or (d) multi-step stateful interactions, the guarded stage can surface runtime semantics and side-effects that static validators cannot anticipate.

Practical recommendations grounded in the artifacts. The archived execution and analysis artifacts suggest three pragmatic paths depending on operator goals: (i) invest in richer static validators that capture more workflow-level constraints when low-latency, high-throughput intent→config translation is the priority; (ii) deploy guarded sandboxing selectively for high-risk or high-impact changes where emulation fidelity matters; (iii) combine both approaches and use adversarial/fuzzing task banks to identify validator gaps—our evidence package includes representative traces that can seed such adversarial test-case design.

Limitations of this interpretation rooted in archived evidence. The near-zero aggregate effect we observed is specific to the preregistered task bank, the single validator implementation, the declared/requested model, and the shared_initial_candidate semantics. The execution manifest and deterministic_reconciliation document the exact run accounting (model_calls_used = 721; repair = 1) and allow independent verification that the observed discordance is localized to a single downstream event. Because the study was powered a priori for a minimum detectable paired difference of 0.20, the empirical observation (≈0.0014) lies far below that sensitivity and must be interpreted as a narrowly scoped near-zero finding for this experimental regime—not as broad evidence that guarded stages are always unnecessary.

Directions where guarded stages are likely to add value. The artifact-grounded diagnostics indicate three concrete extensions where guarded emulation should be retested: (1) adversarially generated tasks specifically designed to exercise validator gaps (e.g., subtle state-dependent ordering errors); (2) expanded vendor-CLI families and syntactic/semantic diversity to challenge parser and validator coverage; (3) higher-fidelity emulation that models vendor-specific control-plane subtleties and long-lived state transitions. The archived per-episode traces in execution/raw_results.jsonl and the representative scoring artifacts provide concrete seed cases and validator-rule examples to guide these follow-up efforts.

Concluding synthesis. The run's provenance and analysis artifacts permit an unambiguous, auditable mapping from the reported aggregate counts to the single discordant episode and the recorded repair-stage call. These artifacts support a principled, evidence-led conclusion: in this preregistered, curated regime, a strong deterministic validator absorbed almost all detectable faults and the guarded sandboxed-emulation stage produced a vanishing marginal detection gain. Whether that conclusion generalizes to more adversarial or production-like regimes requires the targeted extensions described above, for which the current archived bundle provides reproducible starting points.

VI. Limitations

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (gpt-5-mini) and single validator implementation (deterministic_netops_validator_v5) limit generalizability across models and validator families.
- Preregistered shared_initial_candidate semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition

generation variability; this design choice constrains discordance sources to downstream guarded-stage observations or repair calls.
- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in analysis/deterministic_reconciliation.json and provider_call_audit artifacts.

VII. Conclusion

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.001388888888888884) with exact McNemar p = 1.0 and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because shared_initial_candidate semantics were enforced, archived per-episode traces permit independent verification of that mapping via execution/raw_results.jsonl and analysis/deterministic_reconciliation.json. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: execution/execution_manifest.json (execution summary and preregistration SHA-256), execution/raw_results.jsonl (input-results SHA-256 df367ecc40a6f0eaf4021292 60e11d8450884a4ea5fdb94a602e0d2513aef0e8), analysis/deterministic_reconciliation.json (sha256 49731a2e 52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096 d0dec3aa), execution/model_configuration.json (sha256 a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9 ebd073597ffdd820), analysis/paired_contingency_table.csv (sha256 efe6e8e7016fa843a8226e911dbde829 e943903d719101da8d956b3ff899133c), and analysis/condition_summary.csv (sha256 394bc690671b7311 94f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28). The discordant episode is the unique raw_results.jsonl entry with baseline_score = 0 and guarded_score = 1; that record contains the pair_id and validator/emulation traces that map it to the single repair-stage invocation recorded in deterministic_reconciliation.json. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

Disclosure Statement

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as openai_responses). Deterministic validator: deterministic_netops_validator_v5. Orchestration adapter family: hosted_netops_gvr_v1. Preregistration id: f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e (preregistration SHA-256: 0169919fb8c5aedb2c5a78e031f1ab5c3aeda405 cde1ae80fb2f864768595c1b; recorded in the execution manifest). Initial master-prompt immutable reference: provenance/master_prompt.txt, SHA-256 = 1872df1e 1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba. Execution summary (artifact-grounded): completed_episode_count = 1440, complete_pair_count = 720, model_calls_used = 721 (720 shared initial + 1 repair-stage call); stage_counts and provider-call audit are archived in analysis/deterministic_reconciliation.json (sha256 49731a2e52e0...). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

References

[1] Muhammad Bilal, Jon Crowcroft, Ruizhi Wang, Xiaolong Xu, Schahram Dustdar, "Large Language Models for Agentic NetOps and AIOps: Architectures, Evaluation, and Safety", 2026, 10.48550/arxiv.2605.12729

[2] Matt Brown, Ari Fogel, Daniel Halperin, Victor Heorhiadi, Ratul Mahajan, Todd Millstein, "Lessons from the evolution of the Batfish configuration analysis tool", 2023, 10.1145/3603269.3604866

[3] http://arxiv.org/abs/2407.08249